SChimanal

Automated Single-Channel Fluorescence Confocal Image Analysis Workflow for Fiji/ImageJ (1.54p)

Detailed description of the code's functionality (pseudocode)

Implementation note. Thresholding-method consistency was assessed using a Shewhart-inspired criterion rather than a classical Shewhart control chart. For each comparison metric, the median across the nine thresholding methods was used as the central reference value, while the sample standard deviation across the same nine values was used as the measure of dispersion. For particle area and diameter-related metrics, this central value therefore corresponds to the median of the nine model-specific medians. A thresholding method was assigned to the 1 SD, 2 SD, or 3 SD consistency category only when all four evaluated quantities—median particle area, median perimeter-derived diameter, median area-equivalent circular diameter, and particle count—simultaneously satisfied the corresponding criterion.

Particle measurements were performed in two modes: with objects touching the image boundary excluded (boundary-excluded analysis) and with such objects retained (boundary-inclusive analysis). The boundary-inclusive analysis therefore contains all detected particles and is not restricted to particles located at the image boundary.

Spatial units were inherited from the calibration contained in the input microscopy images. Accordingly, particle areas and derived length measurements are expressed in calibrated physical units when valid spatial calibration is present in the source image; otherwise, they remain in pixel-based units.

The software also generates diagnostic plots, segmentation overlays, intermediate data files, and organized output directories to facilitate workflow inspection, quality control, and reproducibility. These diagnostic outputs should be distinguished from the quantitative measurements and summary statistics used for downstream analysis.

Algorithm 1. Automated single-channel fluorescence image analysis

INPUT:
- Confocal fluorescence images
- Set of nine automatic thresholding methods:
  - Default
  - Huang
  - Intermodes
  - IsoData
  - Li
  - MaxEntropy
  - Moments
  - Otsu
  - RenyiEntropy

OUTPUT:
- Particle measurements obtained with each thresholding method
- Particle-size statistics
- Fluorescence-intensity distribution
- Segmentation overlays
- Comparison of thresholding methods

1. Scan the working directory and identify LSM images designated for analysis according to the configured filename convention.

2. FOR each selected input image:
    a. Load the image.
    b. Convert the image to 8-bit format if required.
    c. Select the fluorescence channel:
        IF multiple channels are present:
            split the channels
            identify and select channel 1
            IF channel 1 cannot be identified from its name:
                select the first available channel
        ELSE:
            use the single available channel.
    d. Preprocess the selected fluorescence channel:
        apply Gaussian blur with sigma = 2
        enhance and normalize contrast using 0.35% saturated pixels
        apply smoothing
        apply despeckling.
    e. Save the preprocessed image.
    f. Calculate and save the fluorescence-intensity histogram of the preprocessed image.
    g. Create a working copy of the preprocessed image.
    h. FOR each of the nine automatic thresholding methods:
        i. Duplicate the working image.
        ii. Determine the intensity threshold automatically using the selected thresholding method.
        iii. Convert the thresholded image to a binary mask.
        iv. Apply watershed segmentation to separate adjacent particles.
        v. Perform particle analysis using a lower particle-area limit of 0.1 in the area units recognized by Fiji/ImageJ for the input image. (calibrated area units when spatial calibration is present; otherwise pixel^2).
        vi. Perform boundary-excluded analysis:
            exclude particles touching the image boundary
            measure the remaining particles
            save particle measurements
            save the labelled segmentation overlay.
        vii. Perform boundary-inclusive analysis:
            retain particles touching the image boundary
            measure all detected particles
            save particle measurements
            save the labelled segmentation overlay.
        viii. Close the temporary segmented image.
    END FOR
    i. Calculate particle-size statistics independently for each thresholding method.
    j. Evaluate the consistency of the boundary-excluded particle measurements across the nine thresholding methods using the Shewhart-inspired consistency criterion described in Algorithm 3.
    k. Export particle measurements, summary statistics, fluorescence-intensity data, segmentation overlays, threshold-method comparison outputs, and diagnostic outputs.
END FOR

Algorithm 2. Calculation of particle-size statistics

INPUT:
    Particle measurements obtained for one automatic thresholding method

FOR each detected particle:
    1. Obtain the projected particle area, A.

2. Obtain the particle perimeter, P.
   3. Determine the perimeter-derived diameter: D_P = P / pi
   4. Determine the area-equivalent circular diameter: D_A = 2 * sqrt(A / pi)
END FOR

FOR each thresholding method calculate:
   median particle area
   first quartile (Q1) of particle area
   third quartile (Q3) of particle area
   median perimeter-derived diameter
   Q1 of perimeter-derived diameter
   Q3 of perimeter-derived diameter
   median area-equivalent circular diameter
   Q1 of area-equivalent circular diameter
   Q3 of area-equivalent circular diameter
   total number of detected particles
   square root of the particle count as an auxiliary output quantity.
END FOR

Perform these calculations independently for:
   boundary-excluded analysis
and
   boundary-inclusive analysis.

Export the particle-level measurements and per-method summary statistics.

Algorithm 3. Shewhart-inspired consistency criterion for threshold-method classification and selection

INPUT:
   Boundary-excluded particle statistics obtained independently using the nine automatic thresholding methods

1. For each thresholding method obtain:
      median particle area
      median perimeter-derived diameter
      median area-equivalent circular diameter
      number of detected particles
2. For each of the four quantities calculate the central reference value across the nine thresholding methods:
      reference particle area = median of the nine model-specific median particle areas
      reference perimeter-derived diameter = median of the nine model-specific median perimeter-derived diameters
      reference area-equivalent circular diameter = median of the nine model-specific median area-equivalent circular diameters
      reference particle count = median of the nine model-specific particle counts
3. Calculate the sample standard deviation across the nine thresholding methods separately for:
      median particle area
      median perimeter-derived diameter
      median area-equivalent circular diameter
      particle count
4. FOR each thresholding method:

calculate its absolute deviation from the corresponding across-method median for each of the four quantities.
IF all four quantities are within 1 SD of their respective across-method medians:
classify the thresholding method as "Within 1 SD".
ELSE IF all four quantities are within 2 SD of their respective across-method medians:
classify the thresholding method as "Within 2 SD".
ELSE IF all four quantities are within 3 SD of their respective across-method medians:
classify the thresholding method as "Within 3 SD".
ELSE:
classify the thresholding method as "Outside 3 SD".
END FOR
5. Identify thresholding methods fulfilling the predefined Shewhart-inspired consistency criterion for subsequent quantitative analysis.
6. 6. Export the across-method reference statistics and the thresholding methods classified as Within 1 SD, Within 2 SD, or Within 3 SD, together with their consistency categories.

Algorithm 4. Output generation and organization

INPUT:
Analysis outputs generated for each input image

1. Create a dedicated output directory for each analyzed image.
2. Save the preprocessed image.
3. Save the fluorescence-intensity histogram data and generate a fluorescence-intensity distribution plot.
4. Export fluorescence-intensity data in tabular form.
5. For each thresholding method save:
particle-level measurements
boundary-excluded segmentation overlay
boundary-inclusive particle measurements
boundary-inclusive segmentation overlay.
6. Generate per-method summary statistics for:
particle area
perimeter-derived diameter
area-equivalent circular diameter
particle count.
7. Generate summary tables combining results obtained with the different thresholding methods.
8. Generate a comparison grid containing the preprocessed image and segmentation results obtained using the different automatic thresholding methods.
9. Generate diagnostic plots comparing particle measurements obtained with the different thresholding methods.
10. Export the results of the Shewhart-inspired threshold-method consistency analysis.
11. Organize intermediate and output files into dedicated subdirectories for particle-measurement tables, boundary-inclusive outputs, and Fiji/ImageJ macro files.
12. Complete processing of the current image and proceed to the next designated input image.